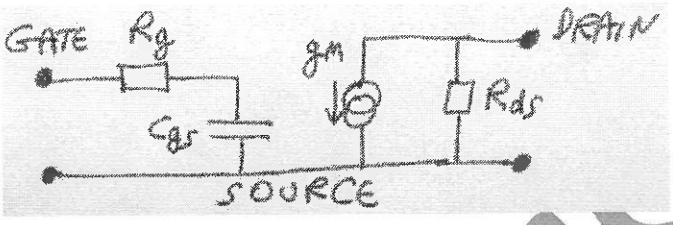
Model answer to Q 1(a): Textbook derivations, computed Examples and Discussions in Class

Consider the simplified equivalent lumped-element model of a MESFET.

i) Draw the equivalent circuit model that only has $R_g = 2\ \Omega, C_{gs} = 50$ pF, $g_m = 100$ mS and $R_{ds} = 50$ kΩ. What can you infer about the signal levels and useful frequency range of this model?



This model only contains fixed (voltage and current invariant) values and so it can only be used for small-signal conditions. Also, it so sparse that it can only be used at very low frequencies, where the parasitic elements available dominate (e.g. below a few MHz).

[2]

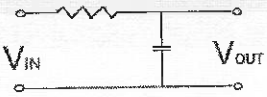
ii) What is the reverse voltage-wave transmission coefficient for the model in 1(a)(i)?

$$S_{12} = 0$$

[1]

iii) For the model in 1(a)(i), connected to an input generator consisting of a voltage source V_S and source impedance $R_S = 50\ \Omega$, derive the input transfer function V_{gs}/V_{IN} and describe the corresponding equivalent circuit model in terms of a filter.

Transfer Functions



Using the ordinary potential divider expression for the 1st order network:

$$H(j\omega) = \frac{V_{OUT}}{V_{IN}} = \frac{1/j\omega C}{R + 1/j\omega C} \qquad \therefore H(j\omega) = \frac{1}{1 + j\omega/\omega_\tau}$$

This transfer function represents the frequency-domain response of the network. This circuit is a low-pass filter, where the voltage is divided by $\sqrt{2}$ at the -3 dB cut-off frequency $\omega_c \equiv \omega_\tau$ for this 1st order network

$$\textit{Relaxation frequency}, \omega_\tau = \frac{1}{\textit{time constant}, \tau = RC}$$

[2]

iv) From the model in 1(a)(iii), calculate the input time constant and relaxation frequency?

$$\text{time constant } \tau = (R_S + R_g)\, C_{gs} = 2.6 \text{ ns}$$

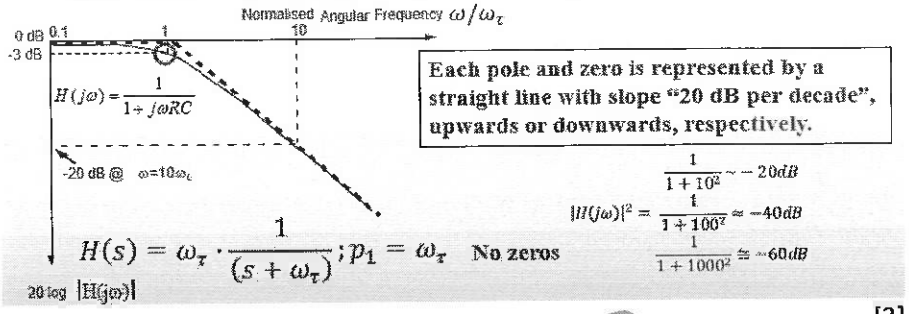
$$\text{relaxation frequency } \omega_\tau = \frac{1}{\tau} = 385 \text{ rad/s} \rightarrow 61 \text{ MHz}$$

[2]

v) With log-log scales, sketch the magnitude of the transfer function obtained in 1(a)(iii) in terms of power response against normalize

frequency between 0.1 and 100. Overlay the corresponding Bode Plot and quantify the frequency behaviour.

- **e.g. 1st order low-pass filter, Bode plot normalising to *RC*:**



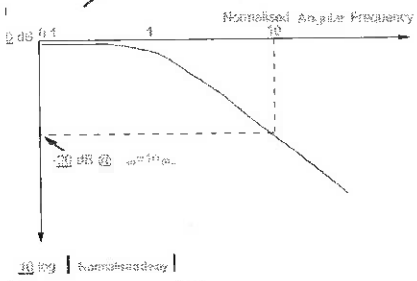
[3]

vi) From the transfer function obtained in 1(a)(iii), derive an expression for the differential-phase group delay. With log-log scales, sketch this function against normalized frequency between 0.1 and 100.

$$H(j\omega)=|H(j\omega)|\,e^{\angle H(j\omega)}$$

Differential-phase group delay (also known as envelope delay):

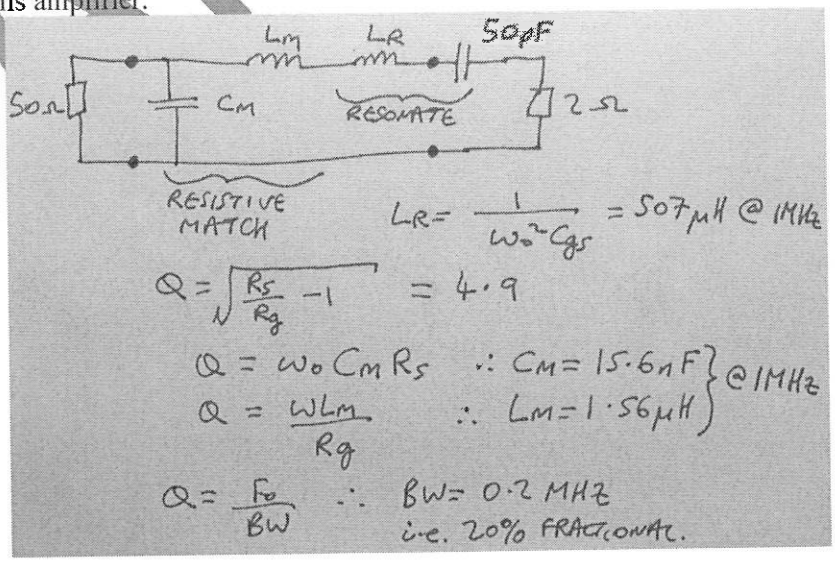
$$\tau(\omega)=-\frac{\partial\angle H(\omega)}{\partial\omega}\rightarrow\frac{\tau}{1+(\omega\tau)^2}$$



[3]

Model answer to Q 1(b): Computed Example

For the MESFET model in 1(a)(i), connected to 50 Ω source and load impedances, design suitable input and output lumped-element matching circuits for operation at 1 MHz and calculate the theoretical bandwidth for this amplifier.



[4]

Model answer to Q 1(c): Discussions in Class

For the MESFET model in 1(a)(i), briefly explain the concepts of AM-to-PM conversion and AM-to-AM conversion. Illustrate the resulting error vectors when both types of conversion effect symbols in an I-Q constellation with a digitally modulated signal.

In practice, C_{gs} is a function of bias and signal voltage levels. Therefore, in a more realistic large-signal MESFET model, C_{gs} can be represented as:

$$Cgs(V) \cong \frac{Cgs(0)}{\sqrt{1-\frac{V_{Cgs}}{\phi}}} \quad where\ \phi = 0.7\,V\ for\ silicon\ and\ 1.3\,V\ for\ GaAs$$

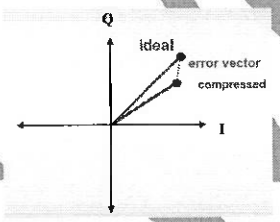
As signal levels increase, C_{gs} decreases and so AM-to-PM conversion occurs.

In practice, g_m is a function of bias and signal voltage levels. Therefore, in a more realistic large-signal MESFET model, g_m can be represented as:

$$g_m = \frac{\partial I_{ds}}{\partial V_{gs}} = \frac{2I_{dss}}{V_p}\left(1-\frac{V_{gs}}{V_p}\right)$$

As signal levels increase, g_m decreases and so AM-to-AM conversion (i.e. gain compression) occurs.

The effects of both conversion processes can be seen on an I-Q constellation plot for a single symbol:



[3]

Model answer to Q 2(a): Computed Example

Consider the satellite communications link illustrated in Figure 2.1. The satellite is in a circular geosynchronous orbit in the plane of the Earth's equator and has an altitude of 35,786 km above sea level. Both the uplink and downlink earth stations are both located on the equator, at sea level. For simplicity, ignore the curvature of the earth.

i) Write down the well-known Friis' range equation for a point-to-point communications link.

$$P_{RX} = P_{TX} \cdot \frac{G_{TX} \cdot G_{RX} \cdot \lambda_o^2}{(4\pi R)^2} \quad Friis'\ Range\ Equation$$

[2]

ii) From 2(a)(i), deduce the expression for the Space-loss term shown in Figure 2.1 and use this to calculate the uplink and downlink ranges.

$$Spaceloss = 10log_{10}\left(\frac{\lambda_o}{4\pi R}\right)^2$$

For 6 GHz uplink: $\lambda_o = 50$ mm and range $R = 39{,}789$ km
For 4 GHz downlink: $\lambda_o = 75$ mm and range $R = 37{,}657$ km

[2]

Answers

iii) What is the separation distance between the uplink and downlink earth stations?

Separation distance is: $\sqrt{(39{,}789\text{ km})^2 - (35{,}786\text{ km})^2} + \sqrt{(37{,}657\text{ km})^2 - (35{,}786\text{ km})^2} = 29{,}115\text{ km}$

[1]

iv) If the uplink and downlink earth station antennas are now pointed directly towards each other and the downlink earth station is retuned to operate at 6 GHz, calculate its LNA gain to achieve the specified output signal power.

Space-loss = -197.3 dB

To achieve no end-to end gain, the LNA gain must now have a signal power gain of:

$$0 - 50 - 60 + 197.3 - 42 - 40 = +5.3\text{ dB}$$

[2]

Model answer to Q 2(b): Textbook Derivation

Consider a signal generator connected to a load impedance, via a transmission line having a power loss L_L, all being matched to the same reference impedance. Assuming the signal generator, transmission line and load impedance are all in thermodynamic equilibrium and at the physical ambient temperature T_o. The input signal power S_i and input noise power N_i to the transmission line can be combined to give an input signal-to-noise power ratio $SNR = S_i/N_i$.

i) Give the mathematical relationship between equivalent noise temperature and noise factor.

Equivalent Noise Temperarture, $T_{RX} = T_o(F_{RX} - 1)$

[1]

ii) Give an expression for the output noise power.

$No = G_{RX}F_{RX}N_i \quad when \quad N_i = kT_oB$

[1]

iii) If the transmission line has a loss of -3 dB, what is its equivalent noise temperature in relation to T_o?

$G = L_L = ½$ and so $F = 1/L_L = 2$ and so $T = 2T_o$

[1]

iv) Determine the output signal power S_o and output noise power N_o from the transmission line and give the output signal-to-noise power ratio $SNR = S_o/N_o$. What is the mathematical relationship between the output and input SNR?

$S_o = S_i/2$ and $N_o = G\,F\,Ni = Ni$ and so $S_o/N_o = (S_i/N_i)/2$. Therefore, $(S_i/N_i)/(S_o/N_o) = F$

[2]

Model answer to Q 2(c): Computed Example

Consider the downlink earth station in Figure 2.1.

i) Write down the well-known Friss' formula for the noise factor of cascaded stages.

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \cdots$$

[1]

Answers

ii) Using the relationship in 2(b)(i) with that in 2(c)(i), derive the equivalent noise temperature for two cascaded stages.

$$T = T_1 + T_2/G_1$$

[2]

iii) If the equivalent noise temperatures for the antenna, LNA and receiver are 38 K, 120 K and 4,000 K, respectively, calculate the system equivalent noise temperature.

$$T_{system} = T_{antenna} + T_{LNA} + \frac{T_{receiver}}{G_{LNA}} = 158.04 \text{ K}$$

[2]

iv) With Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K and a receiver bandwidth of 27 MHz, calculate the noise power at the output of the receiver.

$$N_o = G_{LNA}G_{RECEIVER}(kT_{system}B) = 10^9 \times 58.9 \text{ fW} = 58.9 \text{ μW} = -42.3 \text{ dBW}$$

[1]

v) With the satellite having an EIRP of +34 dBW, calculate the signal power at the output of the receiver.

$$S_o = EIRP - Signalloss + G_{ANTENNA} + G_{LNA} + G_{RECEIVER} = -30 \text{ dBW} = 1 \text{ mW}$$

[1]

vi) Calculate the SNR at the output of the receiver.

$$\frac{S_o}{N_o} = 12.3 \text{ dB}$$

[1]

Model answer to Q 3(a): Bookwork and Discussion in Class

Explain the difference between half-duplex and full-duplex transceiver architectures in how they achieve Tx-Rx isolation and give the name of an international standard for each used in modern mobile communications. Briefly describe the difference between a duplexer, diplexer and triplexer.

* Half-duplex systems (e.g. GSM) only allow the receiver to receive when the transmitter is not transmitting; the transmit and receive frequencies can be identical (e.g. GSM). A high isolation (> 40 dB) transmit/receive (T/R) switch can be used to avoid the receiver being saturated by the transmitter's output signal.

* Full-duplex systems (e.g. CDMA) must use separate transmitter (TX) and receiver (RX) frequencies to avoid the receiver being saturated by the transmitter's output signal. The duplexer and other filters must provide > 100 dB of TX-RX isolation:

A DUPLEXER is a 3-port network that allows the transceiver to use the same antenna. For example, circulator in low-power applications or filters in high isolation applications (e.g. wideband-CDMA).

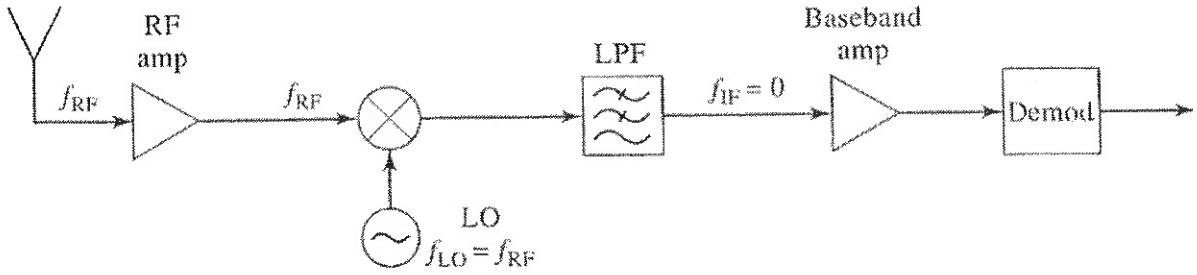
A DIPLEXER is a 3-port filter that either separates or combines signals in 2 different frequency bands. A TRIPLEXER is a 4-port filter used to separate or combine signals in 3 different frequency bands.

[4]

Model answer to Q 3(b): Bookwork and Discussion in Class

Draw the basic block diagram of a Direct Conversion receiver, labelling each block and describe its attributes and drawbacks.

Answers



* Selectivity is controlled by a simple low-pass filter
* Gain is split between the RF and baseband amplifiers
* Since $f_{IF} = 0$, there is no image frequency to deal with (i.e. no Image Rejection filter required)
* High stability (low phase noise and frequency drift) of the LO can be a problem
* Difficult to prevent LO leakage out of the antenna

[10]

Model answer to Q 3(c): Bookwork and Discussion in Class

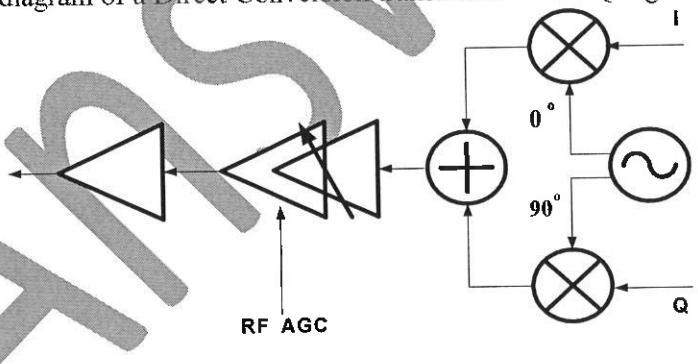
In terms of sidebands, explain when quadrature conversion is required. Give an example of one analogue and one digital modulation scheme that does not require quadrature conversion and one analogue and one digital modulation scheme that requires quadrature conversion.

DSB-SC and BPSK signals have USB = LSB. However, if the USB and LSB contain different information (e.g. QPSK and QAM) then quadrature conversion is required

[2]

Model answer to Q 3(d): Bookwork and Discussion in Class

Draw the basic block diagram of a Direct Conversion transmitter with I-Q digital modulator.

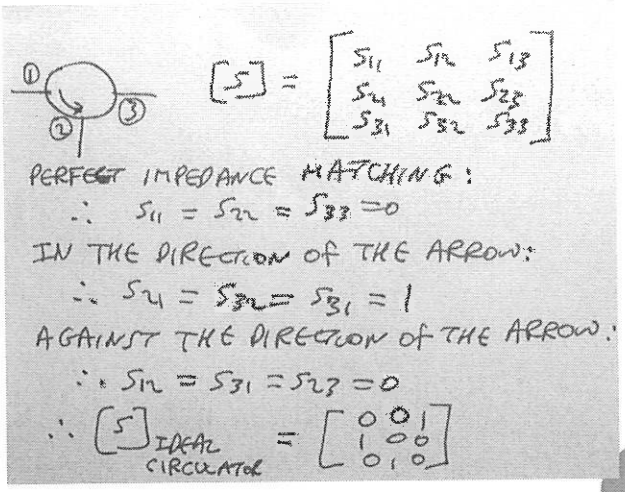


[4]

Model answer to Q 4(a): New application of theory

Draw the conventional symbol for a circulator, indicating each port number, and derive the corresponding S-parameter matrix for an ideal isolator. Clearly explain any assumptions.

Answers

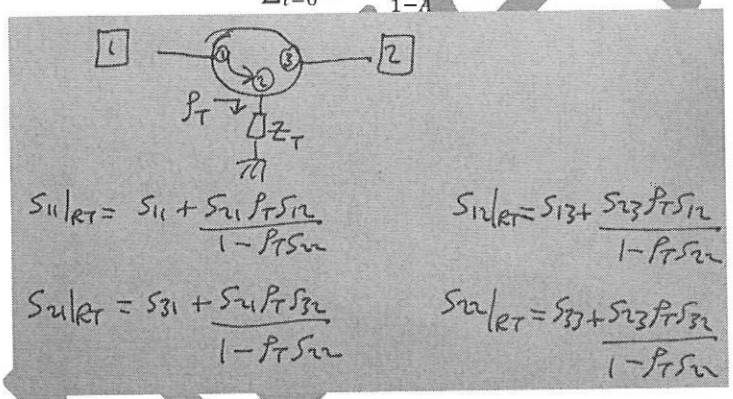


[6]

Model answer to Q 4(b): Discussions in Class

Given a non-ideal circulator, draw a single-stage reflection topology. By inspection, derive expressions for the corresponding S-parameter matrix. The following general mathematical relationship can be used to represent infinite reflections between two impedance boundaries:

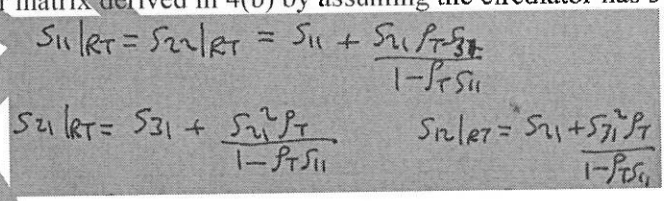
$$\sum_{i=0}^{\infty} A^i = \frac{1}{1-A} \tag{4.1}$$



[5]

Model answer to Q 4(c): Discussions in Class

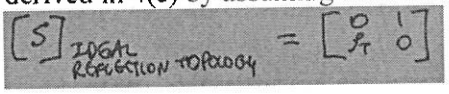
Simply the S-parameter matrix derived in 4(b) by assuming the circulator has 3-way symmetry.



[3]

Model answer to Q 4(d): Discussions in Class

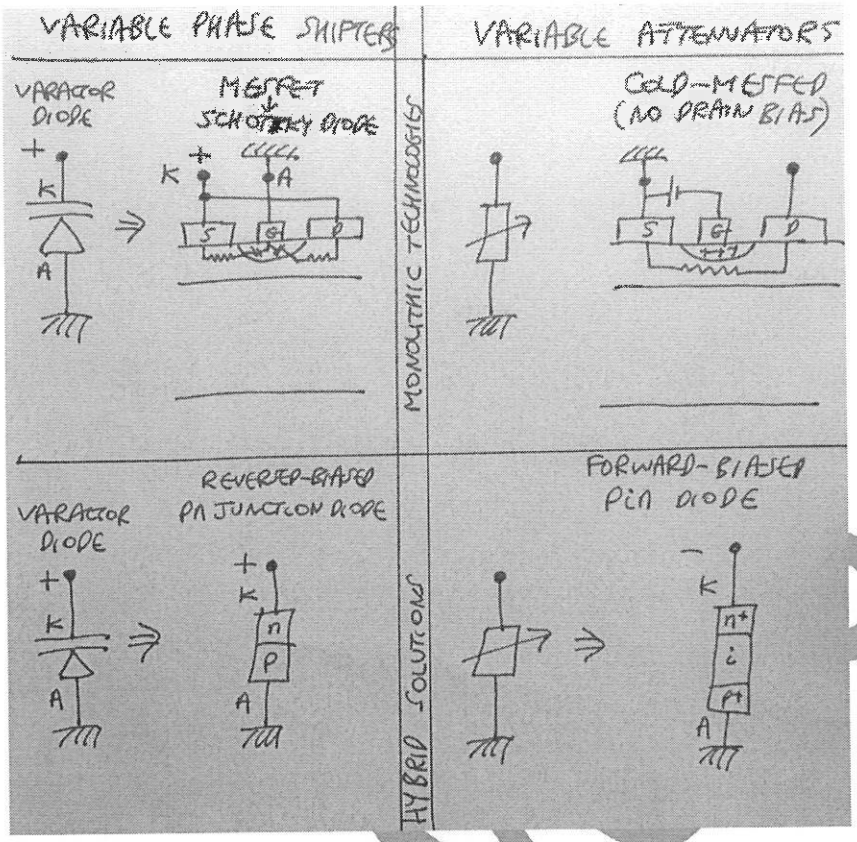
Simply the S-parameter matrix derived in 4(c) by assuming the circulator is ideal.



[2]

Model answer to Q 4(e): Discussions in Class

How can you implement a variable phase shifter and a variable attenuator, using monolithic technology and hybrid solutions?



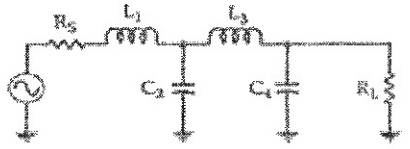
[4]

Model answer to Q 5(a): Computed Example

Design a 4th-order Chebychev band pass filter with 0.5 dB passband ripple, having a centre frequency of 2 GHz and -3 dB bandwidth of 100 MHz, for insertion between 50 Ω source and 99.2 Ω load impedances. What is the theoretical worst-case return loss at centre frequency for this filter?

The low-pass prototype element values are:

n	R_L/R_S	L_1	C_2	L_3	C_4
4	1.984	0.920	2.588	1.304	1.828



- **With low-pass filters, de-normalising is performed directly:**

$$\text{series } L_k = \frac{g_k Z_L}{\omega_c} \qquad C_k = \frac{g_k}{\omega_c Z_L} \text{ shunt}$$

- **With high-pass filters, de-normalising is performed by taking the dual of the LPF:**

$$\text{series } C_k = \frac{1}{g_k \omega_c Z_L} \qquad L_k = \frac{Z_L}{g_k \omega_c} \text{ shunt}$$

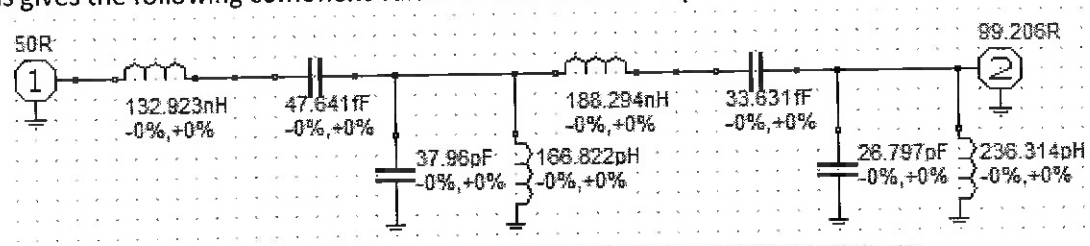
- **With band-pass filters, replace the series element by a series tuned circuit and the shunt element by a parallel tuned circuit, all with $g_k \rightarrow g_k/\Delta$ and $\omega_c \rightarrow \omega_o$:**

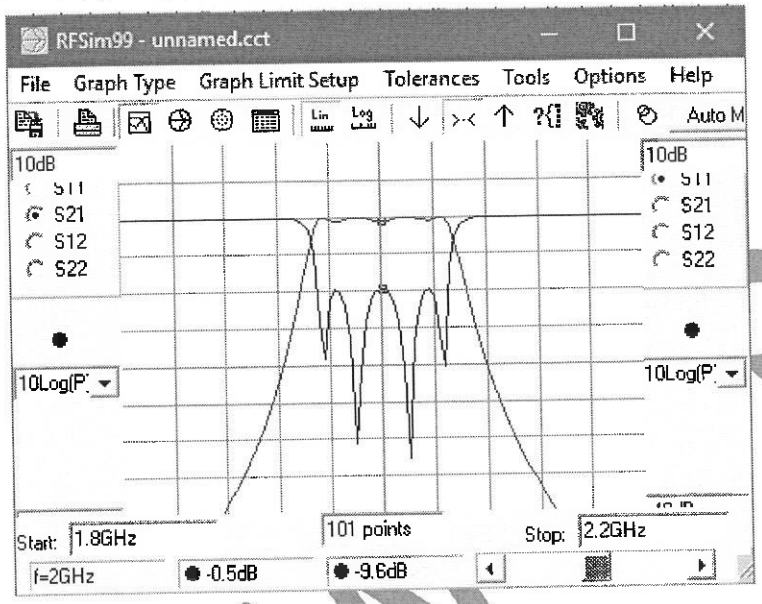
$$\Delta = \frac{\omega_2 - \omega_1}{\omega_o} \equiv \frac{1}{Q_L} \quad \text{fractional -3 dB bandwidth}$$

with respective upper and lower -3 dB cut-off frequencies ω_2 and ω_1

$$\omega_o = \sqrt{\omega_1 \omega_2} \quad \textit{geometric} \text{ mean angular frequency}$$

This gives the following comonent values and simulation response.





S-parameters obey the principle of conservation of energy for a lossless network.

$$1 = |S_{11}|^2 + |S_{21}|^2$$

With -0.5 dB insertion loss $|S_{21}|^2 = 0.891$

$|S_{11}|^2 = 0.1087$ corresponding to a Return Loss $10log_{10}|S_{11}|^2 = -9.64$ dB

[10]

Model answer to Q 5(b): Bookwork and Discussions in Class

Figure 5.1 shows a 2 GHz 'all-transistor' MMIC 3rd-order Chebyshev band pass filter employing three active inductors. List five advantages and five disadvantages to the practical implementation for this filter implementation.

The advantages of active filters are:

1. low passband loss
2. high selectivity
3. easy integration with amplifiers, mixers, oscillators
4. potential for electronic tuning.
5. small size and mass
6. low cost when mass produced

Drawbacks associated with active techniques:

1. poor noise figure
2. non-linearity
3. danger of oscillation
4. complex bias circuitry and significant DC power
5. sensitivity to fabrication tolerances
6. environmental sensitivity

[10]

Model answer to Q 6: Computed Example

i) Calculate the Rollet stability factor and establish the condition for stability.

$$\Delta = -0.99 < 1 \text{ and } K = 0.985 < 1 \text{ conditionally stable!}$$

[3]

ii) From 6(i), as appropriate, calculate either the maximum stable gain or the maximum available gain.

Since $K < 1$, only the maximum stable gain of 100 can be calculated from:

$$\text{MSG} = \left|\frac{S_{21}}{S_{12}}\right|$$

[3]

iii) Assume the transistor has perfect impedance matching at both ports and the output of the transistor is 10 mW with an input DC power of 20 mW. Calculate the input source power, basic efficiency and power-added efficiency.

$$\eta_{add} = \frac{P_{OUT}(fo) - P_{IN}(fo)}{P_{DC}}.100\%$$

$$\therefore \eta_{add} = \eta\left(1 - \frac{1}{G}\right).100\%$$

where basic output efficiency, $\eta = \frac{\text{P}_{\text{OUT}}(fo)}{\text{P}_{\text{DC}}}$ and power gain, $\text{G} = \frac{\text{P}_{\text{OUT}}(fo)}{\text{P}_{\text{IN}}(\text{fo})}$

therefore, η_{add} is always less than η for a finite gain amplifier

$\eta_{\text{add}} \approx \eta$ for high gain amplifiers

$\eta_{\text{add}} = 0$ for unity gain amplifiers

$G_o = |S_{21}|^2 = 100$ and so $P_{in} = 0.1$ mW and $\eta = 50\%$ and $\eta_{ADD} = 49.5\%$

[3]

iv) This transistor is now terminated by source and load impedances that exhibit reflection coefficients of 0.1 and 0.5, respectively. Calculate the source and load impedances, for a system reference impedance of 50 Ω.

$\Gamma = \frac{Z - Zo}{Z + Zo}$ and so $Z = Zo\frac{1+\Gamma}{1-\Gamma}$ and so $Zs = 61.1\ \Omega$ and $Z_L = 150\ \Omega$

[4]

v) With the scenario in 6(iv), calculate the unilateral transducer gain and individual contributions from both the input and out gains.

$$G_{TU} = G_S G_o G_L = 83.83$$

Answers

$$G_S = \frac{1-|\Gamma_S|^2}{|1-S_{11}\Gamma_S|^2} \quad \text{and} \quad G_L = \frac{1-|\Gamma_L|^2}{|1-S_{22}\Gamma_L|^2}$$

$$G_S = 1.01 \text{ and } G_L = 0.83$$

[4]

vi) Using the results from 6(v), which assumes $S_{12} = 0$, calculate the new output power, basic efficiency and power-added efficiency.

With $P_{in} = 0.1$ mW, $P_{out} = 0.1$ mW x 83.83 = 8.383 mW

and $\eta = 41.9\%$ and $\eta_{ADD} = 41.4\%$

[3]

Answers